

11. *nga mara. Kua mutu nga rupu o nga kai i nga paenga o*

All haulm is now stacked at the borders of the plantations.

12. HARATUA. *Kua uru nga kai kai te rua, kua mutu nga mahi a te tangata.*
Crops have now been stored in the store pits. The tasks of man
are finished.⁶

It is important to note that the different names for the Māori months are actually stars, and their heliacal rising in conjunction with the new moon denotes the beginning of the month. In 1854, Mahupuku stated that particular stars were placed in the Milky Way as markers of the various seasons.⁷ Another version of the Māori year that is aligned to the morning rising of stars is as follows:

Te Tahi o Pipiri	The first of Pipiri	Pipiri or Pipirioterangi and the two stars Hamal and Sharatan in the constellation of Aries	(June)
Te Rua o Takurua	The second of Takurua	Takurua is Sirius in Canis Major	(July)
Te Toru Here Pipiri	The third of Toru Here Pipiri	Te Toru Here Pipiri is ζ Perseus	(August)
Te Whā o Mahuru	The fourth of Mahuru	Mahuru is Alpha Hydra	(September)
Te Rima o Kōpū	The fifth of Kōpū	Kōpū is Venus in the morning	(October)
Te Ono o Whitiānaunau	The sixth of Whitiānaunau	Whitiānaunau is γ Leo	(November)
Te Whitu o Hakihea	The seventh of Hakihea	Hakihea is θ Centauri	(December)
Te Waru o Rehua	The eighth of Rehua	Rehua is Antares in Scorpius	(January)
Te Iwa o Rūhiterangi	The ninth of Rūhiterangi	Rūhiterangi is Alniyat in Scorpius	(February)
Te Ngahuru o Poutūterangi	The tenth of Poutūterangi	Poutūterangi is Altair in Aquila	(March)
Te Ngahurumātahi o Paengawhāwhā	The eleventh of Paengawhāwhā	Paengawhāwhā is ε Pegasus	(April)
Te Ngahurumārua o Haki Haratua	The twelfth of Haki Haratua	Haki Haratua is η Pegasus	(May)



Tamatea a Ngana



Tamatea Kai-ariki



Tamatea Tuhāhā



Ōhūa



Atua Whakahaehae



Turu



Ōike



Korekore Tuatahi



Korekore Rawea



Tangaroa Whakapau



Tangaroa Whāriki Koikoi



Ōtane



Ariroa



Rākaunui



Korekore Whakapiri



Ōrongonui



Huna



Rākaumatohi



Tangaroa-ā-mua



Ōmutu



Mawharu



Takirau

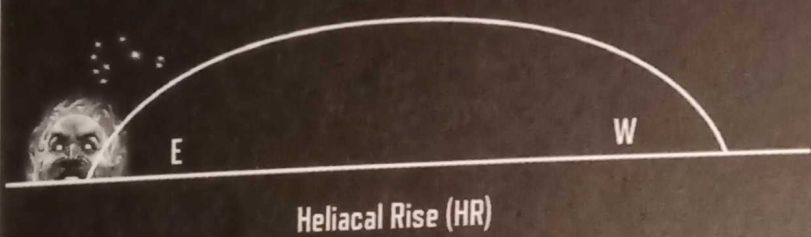


Tangaroa-ā-roto

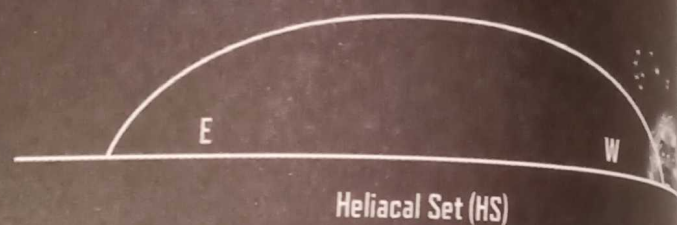


Mutuwhenua

The Moon lunar calendar - Matariki was viewed during the Tangaroa phases, and the Matariki celebration began on Tangaroa-ā-mua and ended at Mutuwhenua. Image: Te Hauwā Tuna 2016



Heliacal Rise (HR)



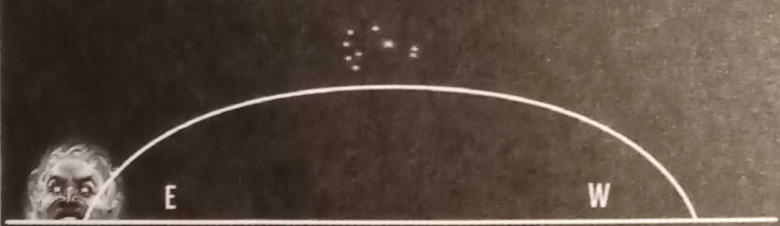
Heliacal Set (HS)



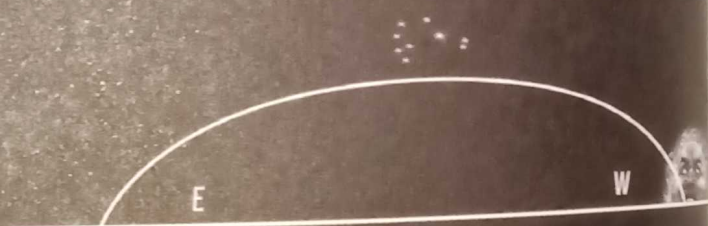
Acronychal Rise (AR)



Acronychal Set (AS)



Dawn Meridional Crossing (Mdawn)



Dusk Meridional Crossing (Mdusk)

Stars were often observed in relation to the position of the sun. This image show how these observations worked.
Image: Te Haunui Tuna 2016.

RISING AND SETTING OF MATARIKI

Ancient cultures throughout the world often viewed stars in correlation with the lunar phase and the rising and setting of the sun, because the combined position of these celestial bodies enabled observers to determine the season.¹ Māori applied this technique when observing stars and, in particular, Matariki.

The image opposite shows the correlation between the position of stars and the rising and setting of the sun; these periods are called heliacal, acronychal and meridional crossing.² The heliacal rise is when a star is seen on the eastern horizon just before the rising of the sun. The heliacal set is when a star is observed on the western horizon just after the sun has set. Acronychal rise is when a star is visible on the eastern horizon as the sun sets in the west, and acronychal set occurs when the star is setting in the west as the sun rises. Dawn meridional crossing is when a star is at its highest point before the sun rises, and dusk meridional crossing is when the star is in the same position after the sun has set. Matariki was mostly observed during its heliacal rising and setting. However, there are also records of it being viewed during its meridional crossing, as the follow passage highlights:

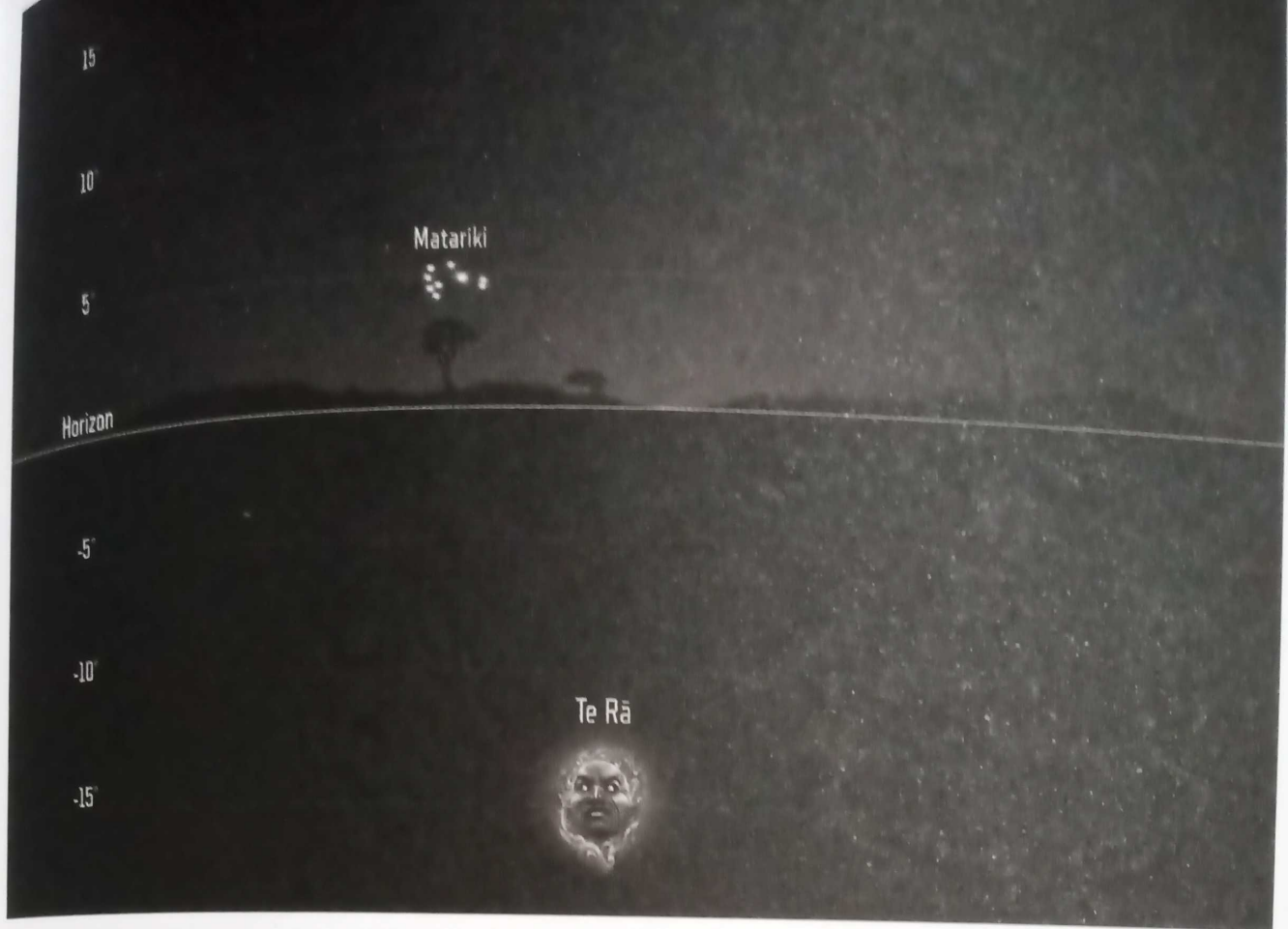
When the rata (*Metrosideros robusta*) was in full bloom, and Mata-riki (Pleiades) had passed the meridian of the sky, and autumn was near, and

when the kumara-crop had been taken up and placed in the storehouses, the young people of Awhitu determined to pay the intended visit to Mount Eden.³

Critical to the observation of the heliacal rising and setting of Matariki is apparent magnitude. In astronomy, stars are given their own apparent magnitude value, a number that astronomers use to measure the brightness of an object in the night sky as seen from earth. The lower the value of the object on the apparent magnitude scale, the brighter it is. Historically, stars were divided into six magnitudes: the brightest became first magnitude stars, and those that are barely visible with the naked eye became sixth magnitude.⁴ Since the invention of the telescope, many more magnitude levels have been added to the measuring system; however, the original magnitude 1 to 6 stars remain.⁵ As already seen, Matariki has an apparent magnitude of 1.6; therefore, it is classed as a third magnitude star.

Object	Māori Name	Apparent Magnitude
Sun	Tamanuiterā	-26.74
Full Moon	Rākaunui	-12.7
Venus	Kōpū	-4.89
Jupiter	Pareārau	-2.94
Sirius	Hinetakurua	-1.46
Canopus	Atutahi	-0.74
Rigel	Puanga	0.13
Antares	Rehua	0.96
Fomalhaut	Ōtamarākau	1.16
Pleiades	Matariki	1.6
Mintaka	Matatoki	2.23

When searching for stars on the eastern horizon before the sun rises, a number of factors become important, including the glow of the sun, the



Matariki at 5 degrees above the horizon with the sun 16 degrees below the horizon, meaning Matariki is visible in the sky.
Image: Te Haunui Tuna 2016.

position of the moon, atmospheric conditions, local topography and the apparent magnitude of the star.

1st magnitude stars become visible at heliacal rise/set at a minimum altitude of 5° ... when the sun has an altitude of -10° . For 2nd magnitude stars, the sun's altitude needs to be a minimum of -14° , and -16° for 3rd magnitude stars.⁶

Since Matariki is a third magnitude star, for its heliacal rise to be visible it will need to be at least 5 degrees above the horizon while the sun is at least 16 degrees below.

Using the Māori lunar calendar, the position of the sun and apparent star magnitude, the following table shows the setting, rising and celebration period for Matariki until 2050.

Matariki

Year	Setting of Matariki	Rising of Matariki	Matariki Period
2017	19 May	17–20 July	17–22 July
2018	7 June	6–9 July	6–13 July
2019	27 May	25–28 June	25 June–3 July
2020	15 May	13–16 July	13–20 July
2021	2 June	2–5 July	2–10 July
2022	23 May	21–24 June	21–29 June
2023	13 May	10–13 July	11–17 July
2024	31 May	29 June–2 July	29 June–6 July
2025	21 May	19–22 June	19–25 June
2026	8 June	8–11 July	8–14 July
2027	29 May	27–30 June	27 June–4 July
2028	16 May	15–18 July	15–21 July
2029	4 June	4–7 July	4–12 July
2030	24 May	23–26 June	23 June–1 July
2031	14 May	11–14 July	11–19 July
2032	1 June	30 June–2 July	30 June–8 July
2033	22 May	20–23 June	20–27 June
2034	11 May	9–12 July	9–15 July
2035	30 May	29 June–1 July	29 June–5 July
2036	18 May	17–20 July	17–22 July
2037	6 June	6–9 July	6–13 July
2038	26 May	25–28 June	25 June–3 July
2039	15 May	13–16 July	13–20 July
2040	2 June	1–4 July	1–9 July
2041	23 May	21–24 July	21–28 July
2042	13 May	10–14 July	10–17 July
2043	1 June	30 June–3 July	30 June–7 July
2044	20 May	19–22 June	19–25 June
2045	8 June	7–10 July	7–14 July
2046	28 May	26–29 June	26 June–4 July
2047	17 May	15–18 July	15–22 July
2048	4 June	3–6 July	3–11 July
2049	24 May	22–25 June	22–30 June
2050	14 May	11–14 July	11–18 July